

Rahul Vashisht

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🌐 <https://vashishtrahul.github.io/>

Research Interests

□ Deep Learning Theory □ Natural Language Processing □ Computer Vision □ Machine Learning
◇ Attention Networks ◇ Learning Dynamics ◇ Gaussian Processes ◇ Interpretability

Education

- 2018 – ··· **Ph.D.** Department of CSE, IIT Madras
Thesis title: *Attention Mechanism: Learning Dynamics and Interpretability*
- 2016 – 2018 **M.Tech. Machine Learning and Computing**, Department of Mathematics, IIST Thiruvananthapuram
Thesis title: *Uncertainty Quantification using Bayesian Networks.*
- 2011 – 2015 **B. Tech Information Technology** GGSIPU, Delhi.
Project title: *Iris Detection based Biometric Authentication.*

Research Publications

Conference Proceedings

- 1 **R. Vashisht** and H. G. Ramaswamy, “Effect of learning rates and parameterisation on interpretability of transformers,” Under Review, Jul. 2025.
- 2 **R. Vashisht** and H. G. Ramaswamy, “On the learning dynamics of attention networks,” in *ECAI 2023*. IOS Press, Sep. 2023, ISBN: 9781643684376. [DOI: 10.3233/faia230541](https://doi.org/10.3233/faia230541).
- 3 L. N. Pandey, **R. Vashisht**, and H. G. Ramaswamy, “On the interpretability of attention networks,” in *Proceedings of The 14th ACML*, ser. Proceedings of Machine Learning Research, vol. 189, PMLR, 2022, pp. 832–847. [URL: https://proceedings.mlr.press/v189/pandey23a.html](https://proceedings.mlr.press/v189/pandey23a.html).
- 4 **R. Vashisht**, H. Viji, T. Sundararajan, D. Mohankumar, and S. Sumitra, “Structural health monitoring of cantilever beam, a case study—using bayesian neural network and deep learning,” in *Structural Integrity Assessment*, Springer Singapore, 2020, pp. 749–761.

Workshop Papers

- 1 R. Mahesh, **R. Vashisht**, and C. Lakshminarayanan, *Transformers with sparse attention for granger causality*, 2024. CODS-COMAD, arXiv: 2411.13264 (cs.LG). [URL: https://arxiv.org/abs/2411.13264](https://arxiv.org/abs/2411.13264).
- 2 **R. Vashisht**, P. K. Kumar, H. V. Govind, and H. G. Ramaswamy, *Impact of label noise on learning complex features*, 2024. SciDL Neurips Workshop, arXiv: 2411.04569 (cs.LG). [URL: https://arxiv.org/abs/2411.04569](https://arxiv.org/abs/2411.04569).
- 3 D. Morwani, **R. Vashisht**, and H. G. Ramaswamy, *Using noise resilience for ranking generalization of deep neural networks*, 2020. Neurips PGDL Competition, arXiv: 2012.08854 (cs.LG). [URL: https://arxiv.org/abs/2012.08854](https://arxiv.org/abs/2012.08854).

Industry Experience

- July 2024 – Dec 2024  **Applied Scientist II Intern**, Amazon, India
Proposed an LLM and Self-Training based model to generate large volumes of pseudo-labels for address matching in geocode learning domain. Improved matcher module performance by using pseudo-labels for downstream tasks such as geocode learning.

Position of Responsibilities

- 2023–2026  **Co-Organizer** Deployable AI Workshop at AAAI Conference
- 2023– · · ·  **Conference duties** ECAI 2023, AAAI 2023–2024, ICLR 2025, IJCAI 2023, 2025, Neurips 2024–2025, AAAI 2026
- 2024–2025  **Instructor Online BS Degree IIT Madras**, Deep Learning for Computer Vision
- 2018 – 2025  **Teaching Assistant, IIT Madras**, Pattern Recognition and Machine Learning, Non-Linear Optimization, Advances in Theory of Deep Learning, Online and Reinforcement Learning,
- 2024  **Intern Mentorship** for project titled, Effects of Label noise on Learning Dynamics in Neural Networks
- 2025  **Intern Mentorship** for project titled: Image Processing Applications for Student Monitoring and Exam Integrity

Tutorials and Invited Talks

- 2024–2025  Tutorial on Transformers and Generative AI model at IIT Madras
- 2023  Oral Presentation at Doctoral Consortium, ECAI-2023
-  Invited Talk at NSE Talent Sprint ADSMI-Cohort at IIT Madras

Awards

- 2022  Best Elevator pitch award for Poster Day, 2022, held at Dept. of CSE, IIT Madras
- 2019, 2021  Received STAR Teaching Assistant award at CSE, IITM
- 2020  Received CoDS-COMAD Student Travel Grant
-  5th Position in Neurips PGDL Competition

Skills

- Languages  Strong reading, writing and speaking competencies for English, Hindi
- Coding  C, C++, Python, Matlab, $\LaTeX$
- Libraries  Pytorch, Tensorflow, ScikitLearn, Theano
- Misc.  Academic research, teaching, workshop organization

References

Available on Request